

Vision–Language Models Read Their Own Attention at the Wrong Depth

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Abstract

Vision–language models decide what to look at — which visual tokens to keep, where to crop — by reading their own attention. We show the field reads it at the wrong depth. Question-conditioned localisation does not exist in the vision encoder (at chance on four architectures), is too weak at early language-model layers to be usable, and becomes reliable only in the upper half of the stack. The practical cost is large and, to our knowledge, unreported: ranking visual tokens for pruning by layer-2 attention — the default in a widely used method — is **worse than ranking them at random** on both models we test, losing 22.0 pp and 13.1 pp against no pruning at all. Reading late layers instead discards **90% of visual tokens at no measurable cost** (−0.5 pp and −1.6 pp), a margin of +24.1 pp and +11.5 pp over the early-layer default, with no training. The same defect appears in crop placement: replacing the standard block-mean read-out with a learned one improves evidence localisation on three of four architectures spanning two model families, and question answering by +12.0 pp and +10.5 pp over an unmodified model. A logit-lens analysis shows why cropping helps at all: it does not improve reasoning but restores an answer-formation step that appears abruptly at a single layer, and only when the crop actually delivers the evidence. We report what does not work with equal weight, including seven internal interventions that all fail and five of our own claims rejected by cross-architecture replication.

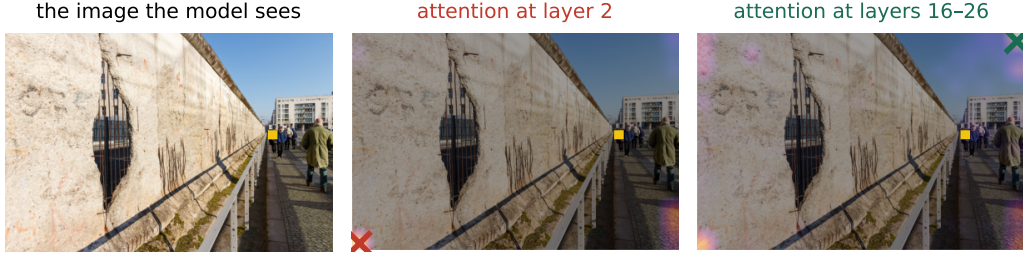
1 Introduction

A vision–language model (VLM) turns an image into a few hundred visual tokens. When the question concerns something small, the evidence occupies a fraction of one token and the model cannot answer: below roughly 0.15 merged tokens of target extent we measure 19.0% accuracy — below the 25% chance level of a four-way question — while the identical question asked of the resolved region is answered correctly 96–100% of the time.

Two large families of methods respond to this, and both make a decision from the model’s attention. Token pruning discards visual tokens judged unimportant. Crop-and-re-encode methods select a region and look again. Both read attention, and the depth at which they read it has been treated as an implementation detail. It is not.

Our contributions:

- Pruning at the wrong depth is worse than not ranking at all. Ranking visual tokens by layer-2 attention scores below random selection on two architectures. A late read-out at the same budget retains the accuracy of using every token while discarding 90% of them (Section 4).
- The standard read-out averages away its own signal. Collapsing a block of layers into one map and taking the maximum is worse than a learned combination on three of four architectures, two model families (Section 3).
- A mechanism, with a causal control. Cropping does not improve reasoning: cropped and uncropped trajectories are indistinguishable for twenty layers and then separate



"What is the color of the woman's scarf?"

Figure 1: The same model, the same forward pass, read at two depths. A real V*Bench item with the queried object boxed in yellow. Middle and right show the model’s actual attention over image positions, upsampled onto the image it received; × is each read-out’s argmax. At layer 2 — where the standard pruning default reads — the mass sits away from the target. At layers 16–26 it concentrates on it. Nothing here is illustrative: both maps come from one inference pass.

Table 1: Evidence localisation, four architectures, two families. “Learned” is a ridge reweighting of per-layer maps, fitted out-of-fold. Chance is $\approx 2.2\%$ throughout.

MODEL	BLOCK MEAN	LEARNED	Δ	95% CI
Qwen3-VL-2B	39.3%	45.5%	+6.3 pp	[+2.6, +10.5]
Qwen2-VL-7B	35.1%	43.5%	+8.4 pp	[+3.1, +13.6]
LLaVA-OneVision-7B	25.1%	26.2%	+1.0 pp	[−2.6, +5.2]
LLaVA-NeXT-7B	12.6%	18.8%	+6.3 pp	[+2.1, +10.5]

abruptly — and the separation appears only on items where the crop actually contains the evidence (Section 5).

- An audited account of failure. Seven internal interventions that would have been free if they worked, all null; and five claims of our own rejected when tested on a second architecture (Section 6, Appendix B).

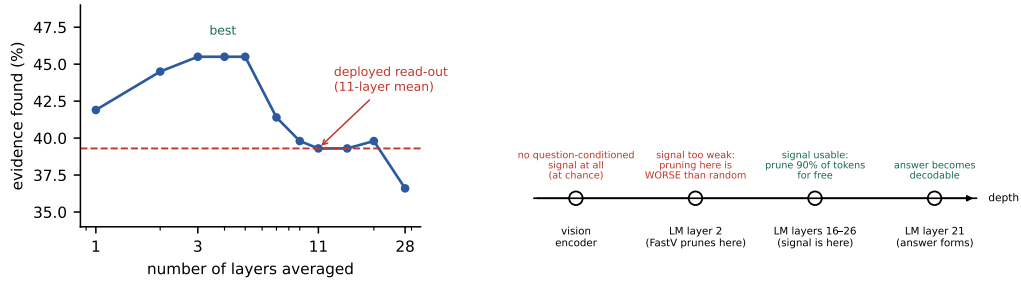
2 Setup

All experiments use V*Bench ($n=191$ four-way questions with ground-truth boxes) and, for transfer, HR-Bench 4k ($n=800$, CircularEval, 4032² images). Models: Qwen3-VL-2B, Qwen2-VL-7B, LLaVA-OneVision-7B, LLaVA-NeXT-7B. Token counts are always measured from the processor’s realised grid, never computed; any arm drifting more than 10% from its target voids its contrast. Every learned component is fitted out-of-fold with folds grouped by item. Appendix A gives the full protocol.

3 The read-out defect

The standard way to locate evidence in a VLM is to average attention over a block of layers and take the argmax. Figure 1 shows why this is lossy: layers disagree about where the target is, and a mean cannot represent disagreement.

Three of four clear zero; LLaVA-OneVision is a null and we report it as one. It is also the only model whose best single layer equals its block mean, i.e. whose layers agree — internally



(a) Averaging the k individually-best layers. More is worse past $k \approx 3$, and the deployed 11-layer mean sits well down the curve. (b) Where the question-conditioned signal lives, and what it costs to read it too early.

Figure 2: The defect (left) and its location in the network (right).

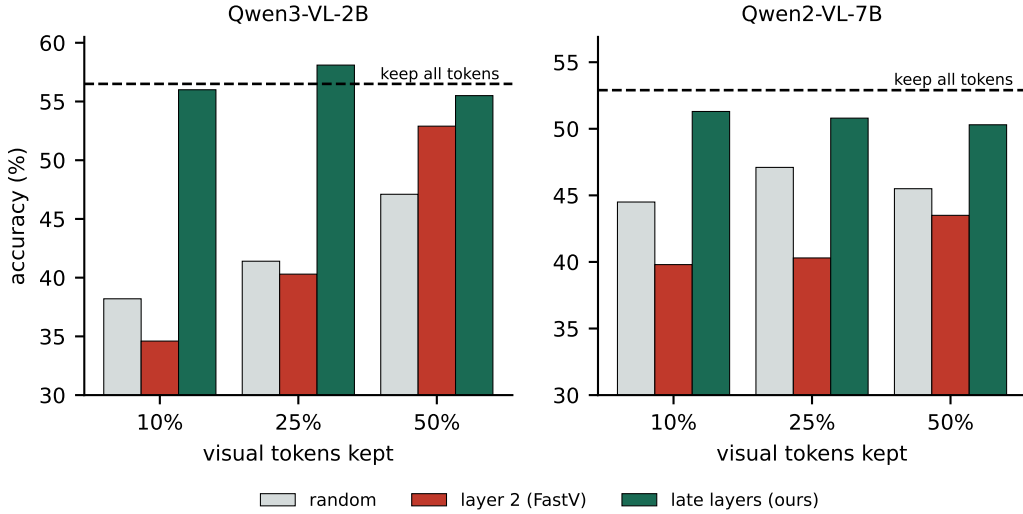


Figure 3: Ranking at layer 2 is worse than ranking at random. Dashed line: keeping every token. A late read-out discards 90% of visual tokens at no measurable cost, on both models.

consistent, but a single data point we do not build on. Absolute localisation varies by a factor of three across models, which matters for anyone porting crop methods to LLaVA.

4 Pruning at the wrong depth

Token pruning ranks visual tokens by attention and drops the low-ranked ones. The widely used default reads that attention at layer 2. Figure 3 shows the consequence. Every arm prunes the same number of tokens, so cost is matched by construction.

At 10% keep, the late read-out is within noise of using every token (-0.5 pp [$-5.8, +4.7$] and -1.6 pp [$-6.3, +3.1$]), while layer-2 ranking loses 22.0 pp [$-29.8, -14.7$] and 13.1 pp respectively and falls below random selection on both models (-3.7 pp, -4.7 pp). The margin is $+24.1$ pp [$+16.2, +31.9$] and $+11.5$ pp [$+4.2, +18.8$].

No learning is required. A plain average of late layers matches a trained signed combination at every operating point ($+2.6, -1.0, +1.0$ pp, all null). The recommendation is a one-line change of which layer index is read.

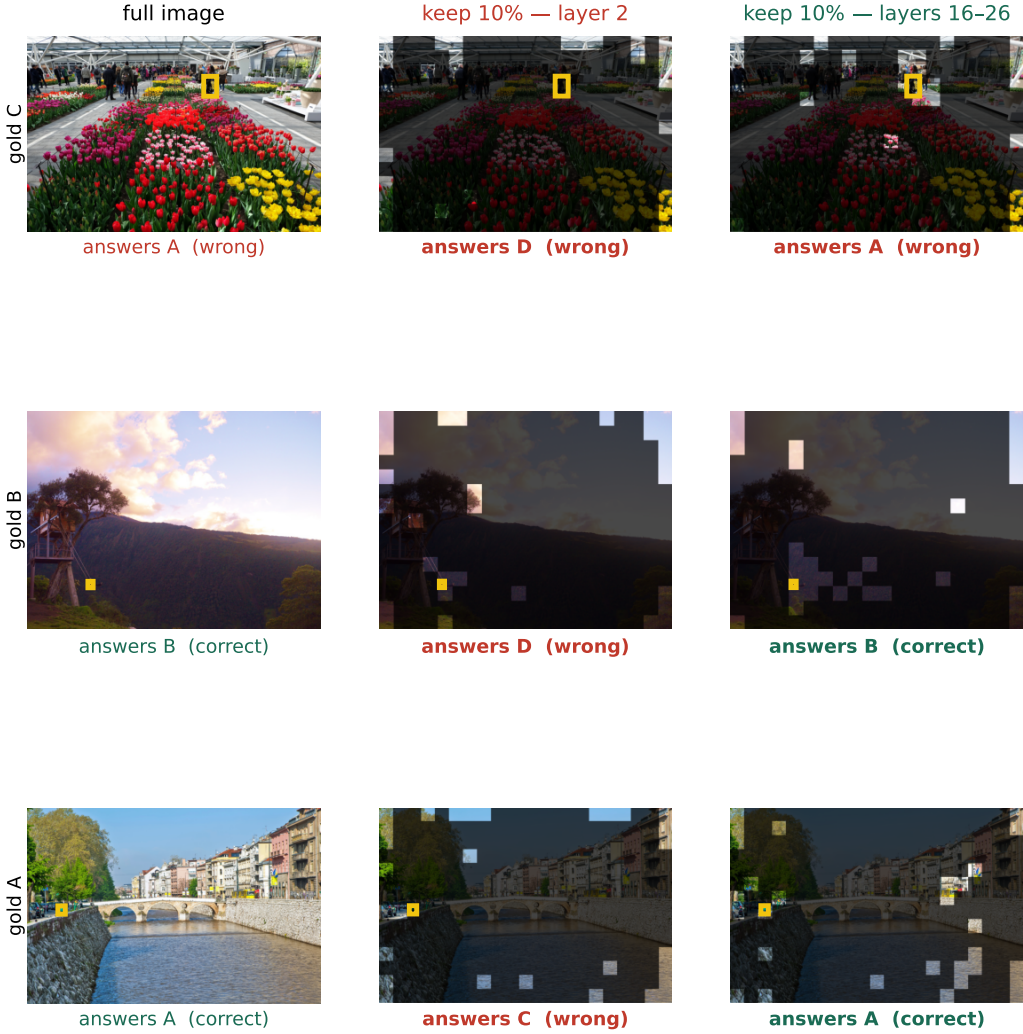


Figure 4: What each ranking actually throws away. Unshaded regions are the 10% of visual tokens kept; everything darkened is discarded before the answer is computed. Yellow box: the queried object. Beneath each panel is the model’s real prediction on that pruned input. Items are the first three in our pool on which the two rankings give different answers — selected on disagreement, never on which ranking turns out to be right, and on the first item both are wrong. Across the full pool of 14, late-layer pruning preserves the full-image answer on 13/14 items and layer-2 pruning on 7/14.

A prediction that failed. We pre-registered that the penalty would be larger on Qwen2-VL, whose early layers rank the target worse. It is smaller (−13.1 vs −21.9 pp). Layer ranking quality therefore does not predict pruning damage: the claim replicates, its causal story does not, and we state them separately rather than let one carry the other.

5 Mechanism: what cropping actually does

Reading the model’s answer at every layer (Figure 5) separates two explanations. A method that improved reasoning would pull ahead early and widen. Instead the cropped and un-cropped arms are indistinguishable through L20, then separate by +15.3 pp at L21 and stay separated. The failing arms jump at L2 — the answer prior forming — and never improve:

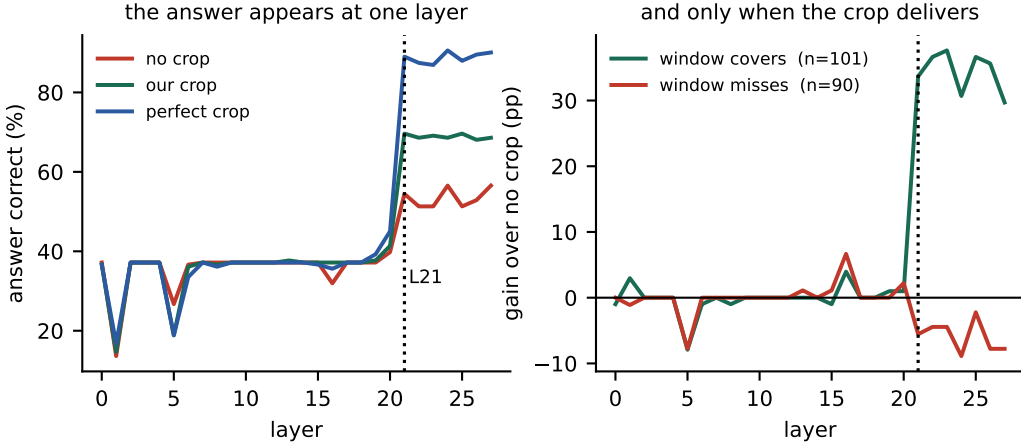


Figure 5: Cropping restores an answer-formation step; it does not improve reasoning. Left: the answer read out at every layer. Right: the same gain split by whether the crop contained the evidence.

the uncropped arm’s best score across all 28 layers equals its final answer, so there is no depth at which the answer was available and lost.

The right panel is the causal control. Splitting the same arm by whether its window contained the evidence: $+29.7$ pp [$+19.8, +39.6$] where it covers, -7.9 pp [$-18.0, +2.2$] where it misses — a 37.6 pp swing. The step appears only when the intervention delivers.

Replication. Both the shape and the control hold on Qwen2-VL-7B. Separation is ≈ 0 through L21 and then steps $(+8, +16, +13, +12, +12, +10$ pp at L22–L27); the index moves by one layer, which we did not require to transfer. The control replicates at $+23.8$ pp [$+13.1, +34.5$] where the window covers against $+0.0$ pp [$-11.2, +11.2$] where it misses, a 23.8 pp swing. One detail does not generalise: on Qwen3-VL a miss is mildly harmful (-7.9 pp) whereas on Qwen2-VL it is exactly neutral, so the cross-model claim is that the gain appears only where the crop delivers, not that a miss actively hurts. The sharpest sub-claim holds identically on both: the uncropped arm’s maximum over all 28 layers equals its final answer (56.3% and 50.8%), so on neither model is there a depth at which the answer was available and lost.

6 What does not work

Seven internal interventions, each free if it worked, are all null: suppressing the serialisation sink, amplifying attention on the target, residual-stream steering, layer read-out selection, DoLa/DeCo, component ablation, and contrastive decoding steered to the ground-truth region (ceiling $+2.6$ pp [$-0.5, +5.8$]). The last is decisive because it aimed perfectly and verified the target is causally live — masking the evidence region shifts logits $3.32\times$ more than masking a random region and costs 10.5 pp — and still could not convert it.

The model looks in the right place; the right place is load-bearing; it still cannot answer, because no operation on the residual stream can supply information the encoder never wrote.

Four attempts to regress a continuous geometric quantity from internals also failed (budget from a perfect size oracle; budget from any free signal; window size from target size; attention-space reallocation). Ranking positions is learnable from internals; regressing scale is not.

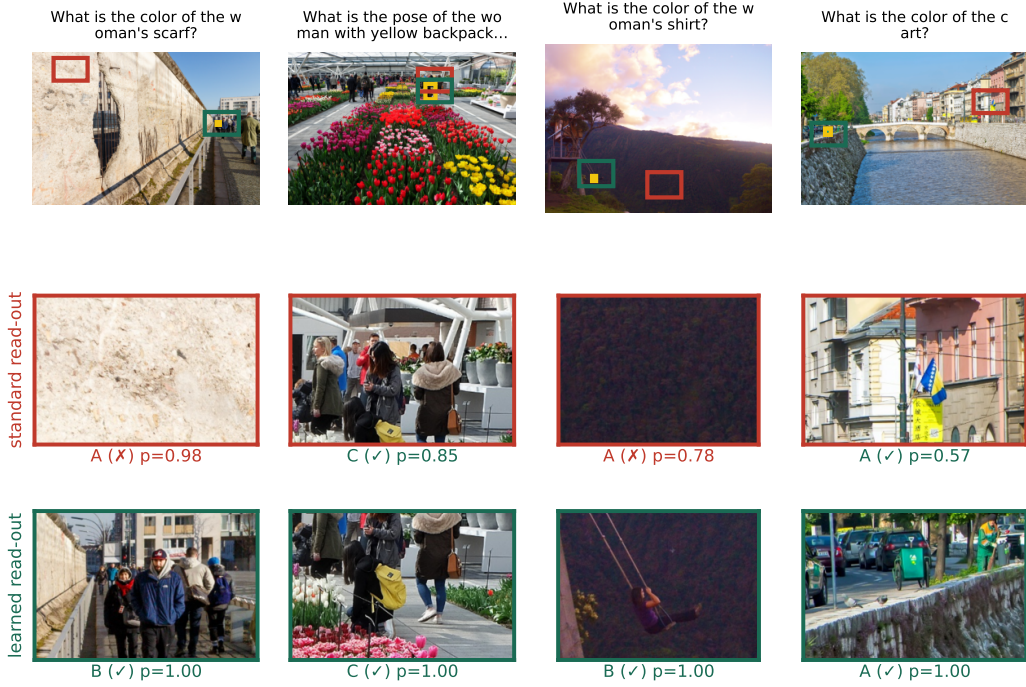


Figure 6: The crops the model is actually fed, and what it actually answers. Top: the source image with the queried object (yellow) and the window each read-out selects. Below: the real cropped input handed to the model, refit to the same 300 tokens, with its prediction and confidence. ✓/× is against the gold answer. Items are the first four in the pool where the standard read-out’s window misses the evidence and the learned one contains it. Note the calibration failure in the leftmost column: the standard read-out crops a featureless stretch of wall and the model answers wrong with $p=0.98$, while the learned crop lands on the people and answers correctly at $p=1.00$. Confidence on a crop that contains nothing is not lower than confidence on a crop that contains the evidence.

7 Limitations

Our crop-placement method improves over an unmodified model on two architectures (+12.0 pp [+4.2, +19.9], +10.5 pp [+2.6, +18.3]) but we have not shown it beats simply spending the same compute on a larger uniform image (+4.7 pp [−3.7, +13.1] and +3.1 pp [−5.2, +11.0]). We report this rather than restricting to the stratum where it clears. Pruning baselines are reimplemented on a common backbone, not run from released checkpoints. Coverage labels require ground-truth boxes and so exist only on V*Bench. Appendix B tags every claim by how many models support it.

A Experimental protocol

A.1 Budgets are measured, never computed

A visual-token budget cannot be set analytically: processors merge patches, pad, and round. Every arm is therefore calibrated by iterating on the processor’s own realised grid until the token count is within tolerance, and the realised count is logged with every record. Any contrast whose arms differ by more than 10% in median realised tokens is voided rather than reported. In the runs reported here all arms land within 0–6.7% of target.

A.2 Numerical precision

Models load in bfloat16. Loading a bf16 checkpoint in fp16 produces all-NaN logits; because `argmax` over NaN returns index 0 and the first option is over-represented in the label distribution, this reads as a clean, plausible, uniformly negative result. It cost us a spurious architecture-specificity finding before it was caught. All scripts now assert `torch.isfinite` on logits and attention before writing any record.

A.3 Cross-validation

Every learned component uses grouped k -fold ($k=5$) with folds grouped by item, so an item’s ~ 300 cells never straddle train and test. Layer-selection results are reported out-of-fold: in-sample selection of a “best layer” is the best of 28 candidates on 191 items and is badly optimistic (Appendix G).

A.4 The logit lens

Intermediate layers are read as `lm_head(norm( $h_i$ ))`; the final layer is taken from `model.logits` and never from `norm( $h_{-1}$ )`. HuggingFace has already applied the final norm to the last hidden state, so re-applying it gives reconstruction error 23.47 against 0.06 — corrupting only that layer, which makes the error invisible in layer-wise plots. Both paths are computed and their disagreement is asserted, so a zero gap now signals that the correct path is not in use.

A.5 Statistics

All intervals are paired bootstrap over items, 10,000 resamples, percentile method. Where a prediction is stated before a run it is written into the script’s docstring with its rejection rule, so a result cannot be reinterpreted after the fact. We report intervals rather than p -values throughout, and we do not correct for multiplicity: the claims that survive are those that replicate on a second architecture, which is a stronger filter than any correction.

B Replication ledger

Our standard is that a claim must hold on more than one model or be rejected rather than caveated. Table 2 is the gate: a claim enters the paper only from the top block.

C Per-layer structure across architectures

Table 3 gives mean `gt_pct` by depth — the rank of the ground-truth cell as a fraction of all cells, where 0.500 is chance by construction and lower is better.

Qwen2-VL’s early and mid layers sit above chance — 15 of its 28 layers are worse than 0.45 — which is why averaging all of them collapses that model to 21.5%. Qwen3-VL’s problem is instead its final layer. The general claim is that anti-correlated layers exist and averaging them in is what hurts; the specific identity of the bad layers does not transfer, and we reject the final-layer claim on that basis (it holds on 1 of 4).

D The allocation ledger: what we could not make work

Four attempts to regress a continuous geometric quantity from model internals, all failed:

- Token budget from target size. With a perfect ground-truth area oracle, the rule $b \geq t^*/\text{area}$ loses to flat uniform in 11 of 12 cells, scored against the uniform ladder at each controller’s own mean spend.
- Token budget from any free pass-1 signal. 17 features (attention concentration, entropy, spread; logit-lens confidence, margin, cross-layer agreement; image geom-

Table 2: Every claim by supporting models.

CLAIM	MODELS	STATUS
Averaging dilutes the read-out	3 of 4, 2 families	survived
Anti-correlated layers exist	4	survived
Layer-2 pruning below random	2	survived
Learned read-out improves proposals	2	survived
Crop method beats unmodified model	2	survived
Answer formation restored by the crop	2	survived
Encoding cliff	2	survived
Serialisation sink	4	survived
“The final layer is anti-correlated”	1 of 4	rejected
“Signed contrast is the mechanism”	1 of 2 tasks	rejected
“No single layer is a good localiser”	false on Qwen2-VL	rejected
Rank-only multi-crop	−3.1 pp vs +5.5 pp predicted	rejected
Crop method beats equal-compute baseline	0 of 2	rejected
Zero-shot transfer to HR-Bench	1	provisional
Evidence region causally live	1	provisional
Seven internal nulls	1	provisional

Table 3: Layer quality by depth. Which layers are anti-correlated differs by architecture; that some are holds on all four.

MODEL	EARLY	MID	LATE	FINAL LAYER
Qwen3-VL-2B	0.456	0.413	0.409	0.529
Qwen2-VL-7B	0.620	0.515	0.392	0.416
LLaVA-OneVision-7B	—	—	—	0.451
LLaVA-NeXT-7B	—	—	—	0.488

etry). Out-of-fold $\rho = 0.254$, but exact-rung accuracy is 9.4% against a 35.6% majority baseline — worse than a constant. The required-budget distribution is bimodal (68/191 items need the smallest budget, 37 need more than the ladder offers) and no free signal makes that call.

- Window size from target size. Across four quartiles spanning a $100\times$ range of target area, the best window is 0.15 in every quartile. A per-item oracle window is worth +11.0pp after a stability check, so the headroom is real — but no predictor we tried reaches it.
- Attention-space reallocation. Amplifying attention on the ground-truth cells recovers 16% of what cropping the same region buys. The target is sub-token: there is nothing to amplify.

Ranking positions is learnable from internals. Regressing scale is not. Every method in this paper that works is a ranking method, and every one that failed asked the model how much rather than where.

E Rank-only multi-crop: a falsified prediction

The ledger above suggests a method that only ranks: take the top- k separated cells, split the budget evenly, show all k in one pass — no size or budget decision anywhere. We priced it before running: union coverage of the top-4 and above the encoding cliff at 75 tokens each is 63.4% against single-crop’s 52.9%, i.e. +10.5pp of answerable items at identical total budget, turning over at $k=5$. At the measured coverage exchange rate this predicts +5.5pp.

Measured: -3.1 pp [-9.4 , $+3.1$]. The prediction had the wrong sign, so the model behind it is wrong, not merely imprecise. The exchange rate was estimated on single crops, where coverage decides whether the evidence is visible at all; with four crops, three are distractors and the account has no term for distractor cost. Ranking still works inside the format — our top-4 beats four random crops by $+27.7$ pp — but the format costs more than the extra coverage buys.

This contradicted an earlier in-project result of $+7.9$ pp for multi-crop, so before recording the negative we tested the clearest difference between the two setups: the connector text between images. Re-running with the descriptive connector gave identical accuracy on all three arms, while 0/25 probability vectors matched byte-for-byte — the prompt changes the output without flipping decisions. The deficit is real, not a prompting artifact.

F Seven internal interventions

Each would have been free if it worked; each is null.

1. Serialisation-sink suppression. Biasing attention away from the trailing row boundary is indistinguishable from biasing away random or interior image tokens (-0.5 pp), while the same operation on prompt tokens costs -17.3 pp — so the machinery bites, and image-token attention is simply not scarce in the direction of removal.
2. Attention amplification on the target. Ground-truth-targeted amplification gains $+5.8$ pp, 16% of what cropping the same region buys. Every practical arm is indistinguishable from a random control.
3. Residual-stream steering. A shared direction exists (cosine 0.53–0.75) and an item’s own displacement is 47% reachable at L20, but the transferable vector gives $+0.0$ pp.
4. Layer read-out selection for the answer. No layer under uniform encoding exceeds 52.9% against the oracle crop’s 92.6%.
5. DoLa / DeCo. Null against a correctly-computed baseline. (Against the double-normalised baseline of Appendix G they appeared favourable, which is how the bug was found.)
6. Component ablation. Null.
7. Contrastive decoding steered to the ground-truth region. The decisive one, because it aims perfectly. Masking the evidence region shifts logits $3.32\times$ more than masking a random region and costs 10.5 pp of accuracy — the region is unambiguously load-bearing — and yet contrast on it reaches a ceiling of $+2.6$ pp [-0.5 , $+5.8$], and -5.5 pp against its own compute-matched bar.

The seventh retires the family. The previous six could each be dismissed as steering to the wrong place; this one steered to the ground truth, verified the place was causally live, and still could not convert it.

G Corrections made during this work

We list these because several changed conclusions, and because the mechanism that caught each one is the part worth reusing.

- A double-normalised logit lens (Appendix A.4) manufactured a “ $+5.1$ pp free read-out gain”, an evidence-dependent late-layer bias, and a favourable comparison against DoLa. Caught because a second code path computed the same quantity and disagreed — 100% agreement at L24, 82.2% at the final layer.
- A transfer result called negative from a prefix. HR-Bench is ordered by category; the first 244 rows were 65% relational, the stratum where our method does nothing. The balanced $n=800$ result reversed the sign ($+0.0 \rightarrow +4.9$ pp, CI clear of zero).

- Two mechanisms proposed for a failure that did not exist. Both were refuted by our own data — “the map is flat at 4K” (peak concentration 0.0228 vs 0.0269, only 1.18 $\times$ apart) and “below the cliff there is nothing to re-rank” (the gain is largest below the cliff).
- A capture fraction against a restricted ceiling. Quoted as 65% of an oracle-among-top-5 gap; against the true ceiling (does any cell cover: 88.5%) it is 27.6%. The loose version also implied the method was near its limit when 35.6 pp of headroom remained.
- An AUROC computed on an unmasked map (0.576); ring-masked as everywhere else in the paper it is 0.745. This briefly made a second signal look competitive when it is substantially worse.
- In-sample layer selection read 41.9% and collapsed to 36.1% out-of-fold — below the block mean it appeared to beat, inverting the conclusion for that model.
- A pre-registered mechanism prediction that failed (Section 4): the layer-2 pruning penalty should have been larger on the model whose early layers rank the target worse. It is smaller. The claim replicates; the causal story does not, and we report them separately.

An executable suite of 86 invariants encodes every item above, so a claim the data stopped supporting fails a test rather than surviving in prose.